Project: Worked examples
Title: Load budget, PSU, transformer and battery
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1. Scope and method

Calculates the demand of a set of loads per supply and the minimum DC power supply, transformer and battery for it. It also gives the heat the loads dissipate for the enclosure calculation.

WHAT IT CALCULATES	RELATION
Demand per load	$P_i \cdot d_i \cdot k_i$; apparent power $P_i \cdot d_i \cdot k_i / \cos\varphi_i$ (arithmetic sum, conservative); this duty-weighted demand sizes the transformer and the enclosure heat
Running load of each load while it is on	$P_i \cdot k_i$, apparent $P_i \cdot k_i / \cos\varphi_i$ (the duty d is the share of time on, so it is left out: a cable, a bus and a power supply carry the load while it is on)
Bus currents	$I_{3\varphi} = S_{3\varphi,run}/(\sqrt{3} \cdot U)$, $I_{1\varphi} = S_{1\varphi,run}/U$, $I_{DC} = P_{DC,run}/U_{DC}$, from the running loads; when a bus is chosen for the DC power supply its input $P_{DC,run}/\eta_{PSU}$ (power factor 1) is added to that bus
Heat dissipated inside the enclosure	$\Phi = \Sigma\, h_i \cdot P_i \cdot d_i \cdot k_i + P_{DC} \cdot (1/\eta_{PSU} - 1)$, h_i = the share of the input power released inside (blank: the loss $1 - \eta_i$; electronics 1; a machine outside the enclosure 0); the second term is the power supply's own loss, present when an AC bus is chosen for it
Power supply sizing	$P_{PSU} \geq SF \cdot P_{DC,run}/f_{derate}$ (the running DC load, duty not applied; f_{derate} from the manufacturer's curve at T_{amb} and altitude)
Transformer sizing	$S_T \geq SF \cdot S$ for the running load; starting peak $S_{peak} = S + \max(m_i \cdot P_i/\cos\varphi_i - S_i) \leq k_T \cdot S_T$ (the starting load undiversified), so $S_T \geq S_{peak}/k_T$ for the start; the rating to buy is the larger of the two
Battery sizing	$C = I_b \cdot t/(DoD \cdot \eta_b \cdot k_{temp} \cdot k_{age})$, $I_b = \Sigma$ backed $P_i \cdot d_i \cdot k_i/(U_b \cdot \eta_{conv})$; Peukert: $I_b \cdot H \cdot (t/H)^{(1/k_p)}$ in place of $I_b \cdot t$, never below $I_b \cdot t$ (the exponent is not valid beyond the rated hour rate H)

Basis: Battery sizing and the Peukert relation[1](Ch. 3–4, 5th ed.); Electrical supply (supply voltage tolerance)[2](cl. 4.3).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Loads	6 row(s)		drawing: E-SCH-010 rev A	any
psu_{bus}	AC bus that feeds the DC power supply (its input P_{DC}/η_{PSU} is added to that bus)	AC single-phase		drawing: E-SCH-010 rev A	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P_{add}	Additional three-phase load (linked from the motor-sizing RMS power)	0	kW	default	0 … 100 kW
$cosphi_{add}$	Power factor of the additional load	0.85		default	0.6 … 1
η_{add}	Efficiency of the additional load	0.9		default	0.6 … 1
$inrush_{add}$	Starting multiple of the additional load	1		default	1 … 10
$heat_{add}$	Share of the additional load's input power released inside the enclosure (a linked machine is outside it: 0)	0		default	0 … 0.5
U_{3ph}	Three-phase line voltage	400	V	spec: Site supply	12 … 690 V
U_{1ph}	Single-phase voltage	230	V	spec: Site supply	12 … 480 V
U_{DC}	DC bus voltage	24	V	drawing: E-SCH-010 rev A	12 … 600 V
T_{amb}	Ambient temperature	45	°C	spec: Plant room design maximum	0 … 60 °C
U_b	Battery nominal voltage	24	V	drawing: E-SCH-010 rev A	2 … 1000 V
η_{psu}	DC power supply efficiency (power factor taken as 1)	0.9		assumption: 24 V supply efficiency 90 %, power factor taken as 1.	0.7 … 1
SF	Sizing safety factor	1.25		assumption: 25 % growth margin.	1 … 1.4
f_{derate}	PSU derating factor at the ambient and altitude	1		assumption: PSU rated to 60 °C without derating (datasheet).	0.1 … 1
k_T	Transformer short-time overload allowed for inrush	3		assumption: Short-time overload allowed by the supply authority.	1 … 20
η_{conv}	Conversion efficiency battery → loads	1		assumption: DC UPS module, loads on the battery bus.	0.7 … 1
DoD	Allowed depth of discharge	0.8		assumption: VRLA, 80 % depth of discharge.	0.1 … 1
η_b	Battery discharge efficiency	0.95		assumption: Discharge efficiency.	0.7 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_{temp}	Temperature capacity factor	1		assumption: Battery kept at 20 °C.	0.3 … 1.2
k_{age}	Ageing (end-of-life) factor	0.8		assumption: End-of-life capacity 80 %.	0.5 … 1
t_{auto}	Battery autonomy	2	h	spec: Telemetry to stay up for 2 h on mains loss	0 … 48 h
P_{PSU}	Selected PSU rating	240	W	drawing: E-SCH-010 rev A	1 … 100000 W
S_T	Selected transformer rating	25	kVA	spec: Existing site transformer allocation	0.01 … 10000 kVA
C_{batt}	Selected battery capacity	17	A·h	drawing: E-SCH-010 rev A	0.1 … 100000 A·h

TABLE ROW	VALUES
loads[1]	name = Duty pump; supply = AC three-phase; P = 11 kW; cosphi = 0.86; eta = 0.91; d = 1; k = 1; inrush = 6.5; backed = no
loads[2]	name = Standby pump; supply = AC three-phase; P = 11 kW; cosphi = 0.86; eta = 0.91; d = 1; k = 0; inrush = 6.5; backed = no
loads[3]	name = Panel fan; supply = AC single-phase; P = 0.06 kW; cosphi = 0.9; eta = 0.7; d = 1; k = 1; inrush = 1; backed = no
loads[4]	name = Panel light; supply = AC single-phase; P = 0.02 kW; cosphi = 0.95; eta = 1; d = 0.1; k = 1; inrush = 1; backed = no
loads[5]	name = PLC and I/O; supply = DC; P = 0.096 kW; cosphi = 1; eta = 0.9; d = 1; k = 1; inrush = 1; backed = yes
loads[6]	name = Router; supply = DC; P = 0.024 kW; cosphi = 1; eta = 0.9; d = 1; k = 1; inrush = 1; backed = yes

3. Assumptions

A1. Each load's P is its electrical input. (Load budget convention.)
A2. DC loads count at power factor 1 in the transformer demand. (Load budget convention.)
A3. With a PSU bus chosen, DC loads count at P/η_{PSU}, which already includes the supply's own loss. (Load budget convention.)
A4. The sheet adds that loss to Φ itself, as $P_{DC}\cdot(1/\eta_{PSU} - 1)$, whenever psu_{bus} and η_{PSU} are set. (Load budget convention.)
A5. Do not add the power supply as a load row, which would count its loss twice. (Load budget convention.)
A6. The heat released inside the enclosure is h·P·d·k per load. (Stated convention for the heat a load releases inside the panel.)
A7. The default h = 1 − η is the loss of a drive or supply. (Stated convention for the heat a load releases inside the panel; change h for a load that differs.)
A8. Electronics release all their input (h = 1) and a machine outside the panel releases none (h = 0). (Stated convention for the heat a load releases inside the panel; change h for a load that differs.)

A9. Apparent powers are added arithmetically, which is conservative when power factors differ. (Upper bound on S.)
A10. PSU derating, transformer overload and battery factors are the manufacturer's values, entered as records. (Entered as reference records, which carry their source, edition and licence.)
A11. η_{psu} (DC power supply efficiency (power factor taken as 1)) is taken as 0.9. (24 V supply efficiency 90 %, power factor taken as 1.)
A12. SF (Sizing safety factor) is taken as 1.25. (25 % growth margin.)
A13. f_{derate} (PSU derating factor at the ambient and altitude) is taken as 1. (PSU rated to 60 °C without derating (datasheet).)
A14. k_T (Transformer short-time overload allowed for inrush) is taken as 3. (Short-time overload allowed by the supply authority.)
A15. η_{conv} (Conversion efficiency battery → loads) is taken as 1. (DC UPS module, loads on the battery bus.)
A16. DoD (Allowed depth of discharge) is taken as 0.8. (VRLA, 80 % depth of discharge.)
A17. η_b (Battery discharge efficiency) is taken as 0.95. (Discharge efficiency.)
A18. k_{temp} (Temperature capacity factor) is taken as 1. (Battery kept at 20 °C.)
A19. k_{age} (Ageing (end-of-life) factor) is taken as 0.8. (End-of-life capacity 80 %.)

4. Working

[22] L1 = P·d·k; P·d·k/cosφ
= 11 × 1 × 1 = 11 kW; 12.791 kVA
= 11 kW
Duty pump (AC three-phase)
[23] L2 = P·d·k; P·d·k/cosφ
= 11 × 1 × 0 = 0 kW; 0 kVA
= 0 kW
Standby pump (AC three-phase)
[24] L3 = P·d·k; P·d·k/cosφ
= 0.06 × 1 × 1 = 0.06 kW; 0.066667 kVA
= 0.06 kW
Panel fan (AC single-phase)
[25] L4 = P·d·k; P·d·k/cosφ
= 0.02 × 0.1 × 1 = 0.002 kW; 0.0021053 kVA
= 0.002 kW
Panel light (AC single-phase)
[26] L5 = P·d·k; P·d·k/cosφ
= 0.096 × 1 × 1 = 0.096 kW; 0.10667 kVA
= 0.096 kW
PLC and I/O (DC)
[27] L6 = P·d·k; P·d·k/cosφ
= 0.024 × 1 × 1 = 0.024 kW; 0.026667 kVA
= 0.024 kW
Router (DC)
[28] $P_{conn} = \Sigma P_i$
= 11 + 11 + 0.06 + 0.02 + 0.096 + 0.024
= 22.2 kW
Connected load
[29] $P_{dem} = \Sigma P_i \cdot d_i \cdot k_i$
= 11 + 0 + 0.06 + 0.002 + 0.096 + 0.024
= 11.182 kW
Demand active power
[30] $S_{dem} = \Sigma P_i \cdot d_i \cdot k_i / \cos\varphi_i$
= 12.791 + 0 + 0.066667 + 0.0021053 + 0.10667 + 0.026667
= 12.993 kVA

Demand apparent power (arithmetic)

[31] $I_{3ph} = S_{3\varphi,run}/(\sqrt{3}\cdot U_{3ph})$, $S_{run} = \Sigma\ P_i\cdot k_i/\cos\varphi_i$

= 12791 / (√3 × 400)

= 18.462 A

Three-phase bus current, loads while on (duty not applied)

[32] $I_{1ph} = S_{1\varphi,run}/U_{1ph}$, $S_{run} = \Sigma\ P_i\cdot k_i/\cos\varphi_i$

= 221.05 / 230

= 0.9611 A

Single-phase bus current, loads while on (duty not applied)

[33] $I_{DC} = P_{DC,run}/U_{DC}$, $P_{DC,run} = \Sigma$ DC $P_i\cdot k_i$

= 120 / 24

= 5 A

DC bus current, loads while on (duty not applied)

[34] $\Phi = \Sigma\ \dot{q}\cdot P_i\cdot d_i\cdot k_i + P_{DC}\cdot(1/\eta_{PSU} - 1)$

= 0.09×11 + 0.09×0 + 0.3×0.06 + 0×0.002 + 0.1×0.096 + 0.1×0.024 + PSU loss 0.013333

= 1.0333 kW

Heat dissipated inside the enclosure (→ enclosure-heat P_{add})

[35] $P_{DC} = \Sigma$ DC $P_i\cdot d_i\cdot k_i$

= 120 W

= 120 W

DC demand (duty-weighted)

[36] $P_{DC,run} = \Sigma$ DC $P_i\cdot k_i$

= 120 W

= 120 W

DC load while on (duty not applied)

[37] $P_{PSU,req} = SF * P_{DC,run} / f_{derate}$

= 1.25 × 120 W / 1

= 0.15 kW

Minimum PSU rating (the supply carries the load while it is on)

[38] $S_{T,req} = SF * S_{dem}$

= 1.25 × 12.993 kVA

= 16.241 kVA

Minimum transformer rating for the running load

[39] $S_{peak} = S_{dem} + \max(m_i\cdot P_i/\cos\varphi_i - S_i)$

= 12.993 + 70.349 (Duty pump starting)

= 83.342 kVA

Peak apparent power while the largest load starts

[40] $S_{T,req,start} = S_{peak} / k_T$

= 83.342 kVA / 3

= 27.781 kVA

Minimum transformer rating for the starting peak

[41] $S_{T,min} = \max(S_{T,req}, S_{T,req,start})$

= max(16.241 kVA, 27.781 kVA)

= 27.781 kVA

Minimum transformer rating (running load and starting peak)

[42] $I_b = \Sigma$ backed $P_i\cdot d_i\cdot k_i/(U_b\cdot\eta_{conv})$

= 120 / (24 × 1)

= 5 A

Battery discharge current

[43] $Ah = I_b * t_{auto}$

= 5 A × 2 h

= 10 A·h

Discharge capacity

[44] C_{req} = Ah / (DoD * η_b * k_{temp} * k_{age})
= 10 A·h / (0.8 × 0.95 × 1 × 0.8)
= 16.447 A·h
Minimum battery capacity
[45] U_L = U_{3ph}
= 400 V
= 400 V
Three-phase line voltage of the bus
[46] pf_{link} = cosφ of the linked machine
= 0.85
= 0.85
Power factor of the linked machine
[48] u_{PSU} = $P_{PSU,req}$ / P_{PSU}
= 0.15 kW / 240 W
= 0.625
PSU utilisation
[50] u_T = $S_{T,req}$ / S_T
= 16.241 kVA / 25 kVA
= 0.64964
Transformer utilisation
[51] $S_{T,surge}$ = k_T * S_T
= 3 × 25 kVA
= 75 kVA
Short-time rating of the selected transformer for the starting peak
[52] u_{inrush} = S_{peak} / $S_{T,surge}$
= 83.342 kVA / 75 kVA
= 1.1112
Inrush utilisation
[54] u_{batt} = C_{req} / C_{batt}
= 16.447 A·h / 17 A·h
= 0.96749
Battery utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
P_{dem}	11.18 kW	Demand active power
S_{dem}	12.99 kVA	Demand apparent power
I_{3ph}	18.46 A	Three-phase bus current
I_{1ph}	0.9611 A	Single-phase bus current
I_{DC}	5 A	DC bus current
Φ	1.033 kW	Heat dissipated inside the enclosure
U_L	400 V	Three-phase line voltage
pf_{link}	0.85	Power factor of the linked machine
C_{req}	16.45 A·h	Minimum battery capacity
P_{conn}	22.20 kW	Connected load
$P_{PSU,req}$	0.1500 kW	Minimum PSU rating
$S_{T,req}$	16.24 kVA	Minimum transformer rating, running load

SYMBOL	VALUE	DESCRIPTION
S_{peak}	83.35 kVA	Peak apparent power at start (governing)
$S_{T,req,start}$	27.78 kVA	Minimum transformer rating, starting peak
$S_{T,min}$	27.78 kVA	Minimum transformer rating, running and starting
I_b	5 A	Battery discharge current
Ah	10 A·h	Discharge capacity
u_{PSU}	0.62	PSU utilisation
u_T	0.65	Transformer utilisation
u_{inrush}	1.12	Inrush utilisation
u_{batt}	0.97	Battery utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Starting peak against the transformer short-time rating	83.35 kVA	75 kVA	1.12	
PASS	Required battery capacity against the selected capacity	16.45 A·h	17 A·h	0.97	
PASS	Required transformer rating against the selected rating	16.24 kVA	25 kVA	0.65	
PASS	Required PSU rating against the selected rating	0.1500 kW	240 W	0.62	

7. Applicability

WARNING records: tabulated values not taken from a reference record are f_{derate} (PSU derating); DoD, η_b, k_{temp}, k_{age} (Battery factors). Enter each as one so its source, edition and licence travel with it. [record]

WARNING f_{derate}: the ambient is above 40 °C but no PSU derating is entered. Read the factor from the manufacturer's curve. [derate]

WARNING t_{auto}: the power supply rating covers the DC loads only, not the battery's charging power. Check that the supply can charge the battery as well as carry the load. [battery-charge]

Defaults used (inputs left empty):

P_{add}: not entered, so the default of 0 kW is used.

$cosphi_{add}$: not entered, so the default of 0.85 is used.

η_{add}: not entered, so the default of 0.9 is used.

$inrush_{add}$: not entered, so the default of 1 is used.

$heat_{add}$: not entered, so the default of 0 is used.

8. Verification

```
Template elec.load-budget@1.6.1 Content hash 37b6289e618acdb2 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Reddy (ed.), Linden's Handbook of Batteries.
2. BS EN 60204-1:2018.